

■■■ OpsMind AI

AI Powered Operational Management System with Predictive Intelligence & Autonomous Decision Support

Project Title: OpsMind AI	Version: 1.0.0 Production
Author / Lead: Operational Engineering Team	Status: Fully Implemented & Deployed
Repository: GitHub / Ragavimurugesh / opsmindai	Database: SQLite (12 Tables Seeded)

1. Executive Summary & Problem Statement

Modern enterprises face major operational challenges including bottleneck delays, resource underutilization, unforeseen workflow risks, and delayed decision-making. Conventional operations tools function strictly as passive databases without offering proactive risk forecasting or predictive suggestions.

OpsMind AI solves this problem by combining **Artificial Intelligence**, **Predictive Analytics**, **Digital Twin Simulation**, and **Autonomous Decision Support**. The system dynamically predicts operational risks, simulates scenario changes, flags abnormal behaviors, and gives managers real-time actionable recommendations.

2. System Objectives & Key Innovations

- ■ **Digital Twin Simulation:** Virtual environment allowing managers to run 'what-if' resource allocation scenarios before committing physical assets.
- ■ **AI Risk Prediction:** Machine learning models forecasting operational completion probabilities, efficiency scores, and potential delays.
- ■ **Autonomous Decision Support:** Auto-generated actionable recommendations for load balancing, cost optimization, and preventative maintenance.
- ■ **Operational Health Index:** Real-time 0–100 composite scoring dashboard assessing overall enterprise health.
- ■ **Anomaly Detection Engine:** Continuous monitoring algorithm detecting unusual resource usage spikes and schedule delays.
- ■ **Generative AI Assistant:** Intelligent conversational chatbot providing real-time operational query responses.

3. System Architecture & Tech Stack

Layer	Technologies	Purpose & Details
Frontend UI	React 18, React Router v6, Axios, Custom Dark CSS	Interactive dashboard, telemetry widgets, glassmorphism UI layout
Backend API	Python 3.11, FastAPI, Uvicorn, Pydantic	RESTful micro-service API endpoints, CORS support, router modules
Machine Learning	Scikit-Learn, RandomForest, NumPy, Pandas	Predictive modeling for performance & risk scoring
Database Layer	SQLite 3, SQLAlchemy ORM	12 relational tables storing operational entities and system logs

4. Database Architecture (12 Relational Tables)

Table Name	Primary Key	Description & Attributes
Users	user_id	User credentials, emails, passwords, and RBAC roles (Admin, Manager, Employee)
Operations	operation_id	Operational workflows, status (Pending/InProgress/Completed), performance scores
Resources	resource_id	Hardware/Cloud assets, utilization rates (%), and availability statuses
Digital_Twin	twin_id	Simulation scenarios, virtual environment parameters, and outcome outputs
Predictions	prediction_id	ML prediction outputs, target metrics, prediction dates, and model names
Risk_Analysis	risk_id	Risk level classifications (Low/Medium/High/Critical) and risk scores
Recommendations	recommendation_id	Autonomous AI suggestions, priority flags, and action dates
Anomaly_Detection	anomaly_id	Detected anomalies, severity flags, and automated system alerts
Health_Score	score_id	Efficiency score, productivity score, and overall composite score
Reports	report_id	Generated operational executive reports and analytics content
Notifications	notification_id	Event-driven system alerts, unread states, and timestamp logs
AI_Assistant_Logs	log_id	Conversational history, user queries, and AI assistant response logs

5. Verification & Deployment Results

- ✓ **Backend Server Health:** FastAPI running on http://127.0.0.1:8000 with CORS enabled.
- ✓ **Frontend Build Status:** React application compiled successfully into optimized production build.
- ✓ **Database Seeding:** Automated seed_data.py script populated 12 database tables with 80+ records.
- ✓ **API Telemetry Test:** All 13 module endpoints responding with **HTTP 200 OK**.
- ✓ **GitHub Repository:** Code committed and pushed to <https://github.com/Ragavimurugesh/ai-powered-operational-management-system>.

End of Report — OpsMind AI System Documentation